

Cloud Computing

Cloud computing is the use of various services, such as software development platforms, servers, storage, and software, over the Internet, often referred to as the "cloud". It is defined as a type of computing that relies on sharing computing resources rather than having handle applications. In cloud computing, the word cloud is used to represent "the Internet," so the phrase cloud computing means "a type of Internet-based computing," where different services - such as servers, storage, and applications are delivered to an organization's computers and devices through the Internet.

Cloud computing allows application software to be operated using internet-enabled devices.

Types of Clouds

Clouds can be classified as public, private, and hybrid. Public cloud is made available to the general public or a large industry group. Private cloud computing environment resides within the boundaries of an organization and is used exclusively for the organizational benefits. Hybrid cloud is the combination of both public and private cloud. Sensitive With this cloud organizations might run non-core applications in a public cloud, while maintaining core applications and data in a private cloud.

Service Models of Cloud Computing

1. IaaS (Infrastructure as a Service): In this service, computing infrastructural components like server hardware, storage, bandwidth, and other fundamental computing resources are provided through the cloud.

2. SaaS (Software as a Service): This service includes complete software on the can access software hosted on the cloud without installing it on the user's own computer.

3. PaaS (Platform as a Service): It allows the user to rent virtualized servers and associated services used to run existing applications, or to design, develop, test, deploy and host applications. It provides clients with access to the basic operating software and optional services to develop and use software applications without the need to buy and manage the underlying computing infrastructure.

Advantages of Cloud Computing

Some of the advantages of this technology are:

1. Cost-efficient: It is probably the most efficient method to use, maintain and upgrade.
2. Almost unlimited storage: Storing information in the cloud gives us almost unlimited storage capacity.
3. Backup and recovery: Since, all the data is stored in the cloud, backing it up and restoring the same is relatively much easier than storing the same on a physical device.
4. Automatic software integration: In the cloud, software integration is usually something that occurs automatically. It also allows us to customize the options with great ease.
5. Easy access to information: Once the user is registered in the cloud, the user can access the information from anywhere, where there is an Internet connection.

6. Quick deployment: Once the method of functioning is selected, the entire system can be fully functional in a matter of few minutes.

Disadvantages of Cloud Computing

Despite its many benefits, as mentioned above, cloud computing also has its disadvantages.

1. Technical issues: This technology is always prone to outages and other technical issues. Even the best cloud service providers run into this kind of trouble. Despite keeping up high standards of maintenance.
2. Security in the cloud: Storing all the sensitive information to a third-party cloud service provider could potentially put the company at great risk.
3. Prone to Attack: Storing information in the cloud could make the company vulnerable to external threats and attacks.

Big Data:

Big Data refers to complex and large data sets that have to be processed and analyzed to uncover valuable information that can benefit businesses and organizations.

It has features like:

1. It refers to a massive amount of data that keeps on growing exponentially with time.
2. It is so voluminous that it cannot be processed or analyzed using conventional data processing techniques.
3. It includes data mining, data storage, data analysis, data sharing, and data visualization.
4. The term is an all-comprehensive one including data, data frameworks, along the tools and techniques used to process and analyze the data.

According to Gartner, the definition of Big Data- "Big data is high-volume, velocity, and information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making."

Types of Big Data

Big data can be classified as Structured, unstructured, and semi-structured.

1. Structured: It means that data can be processed, stored, and retrieved in a fixed format. It refers to highly organized information that can be readily and seamlessly stored and accessed from a database by simple search engine algorithms.
2. Unstructured: It refers to the data that lacks any specific form or structure whatsoever. This makes it very difficult and time-consuming to process and analyze unstructured data.
3. Semi-structured: It relates to the data containing both the formats mentioned above, that is structured and unstructured data. To be precise, it refers to the data that although has not been classified under a particular repository (database), yet contains vital information or tags that segregate individual elements within the data.

Characteristics of Big Data

The main characteristics of big data are:

1. Variety: It refers to the variety of data gathered from multiple sources. The variety can be structured, unstructured, or semi-structured.

2. Velocity: It refers to the speed at which data is being created in real-time. It also comprises the rate of change, linking of incoming data sets at varying speeds, and activity bursts.
3. Volume: Big Data indicates huge 'volumes of data that are being generated daily from various sources like social media platforms, business processes, machines, networks, human interactions, etc.
4. Veracity: It refers to the reliability or trustworthiness of the data. Due to the large volume of data, we have uncertainty about the validity, the accurateness of data.
5. Value: It refers to the worth of business value of the collected data.
6. Variability: It refers to the inconsistency of the big data and how the big data can be used and formatted.

Application Areas of Big Data

Major application of big is data is:

1. Healthcare or Medical sector.
2. Academia.
3. Banking.
4. Manufacturing.
5. Information Technology (IT).
6. Retail business.
7. Transportation.

Advantages of Big Data Processing

Some of the advantages of big data processing are:

1. Businesses can utilize outside intelligence while taking decisions.
2. Improved customer service.
3. Early identification of risk to the product/services,
4. Better operational efficiency.
5. Big data analysis derives innovative solutions. It helps in understanding and targeting customers. It helps in optimizing business processes.

Disadvantages of Big Data Processing

Despite its many benefits, big data processing has the following disadvantages.

1. Traditional storage can cost a lot of money to store big data.
2. Big data analysis is not useful in the short run. It needs to be analyzed for a longer duration to leverage its benefits.
3. Big data analysis results are sometimes misleading.